

ST BENEDICT'S COLLEGE



SUBJECT	AP MATHEMATICS	DATE	22 July 2015
GRADE	11	MARKS	200
EXAMINER	Mr Benecke	MODERATOR	Mrs. Povall and Mr. Gill
NAME	Memo	DURATION	2 hours + 30 min
CLASS			

QUESTION NUMBER	DESCRIPTION	MAXIMUM MARK	ACTUAL MARK
1	Equations with Fractions	26	
2	Rational Inequalities	28	
3	Equations with complex roots	12	
4	Absolute Values	29	
5	Trigonometry	24	
6	Graphs and Split Graphs	14	
7	Venn Diagrams	31	
8	Conditional Probability	20	
9	Counting Methods	16	
	Total	200	

1. This paper consists of 22 pages
2. Write your answers on this paper.
3. Folio paper is provided for rough work and won't be marked, if you would like your work on the folio paper to be marked indicate this clearly.
4. Round off all answers to **2 decimal places** unless otherwise stated
5. Calculators may be used.

QUESTION 1 - SOLVE FOR x

26 MARKS

NOTE: ($x \in \mathbb{R}$)

a) $\frac{x+6}{x^2-9} - \frac{x-9}{(x-1)(3-x)} = \frac{2x-1}{x^2+2x-3}$ (8)

$$\frac{x+6}{(x-3)(x+3)} \checkmark - \frac{x-9}{(x-1)(3-x)} = \frac{2x-1}{(x+3)(x-1)} \checkmark$$

Rest
 $x \neq \pm 3$ ✓
 $x \neq 1$ ✓

$$\frac{x+6}{(x-3)(x+3)} + \checkmark \frac{x-9}{(x-1)(x-3)} = \frac{2x-1}{(x+3)(x-1)}$$

LCD = $(x-3)(x+3)(x-1)$ ✓

$$(x+6)(x-1) + (x-9)(x+3) = (2x-1)(x-3) \checkmark$$

$$x^2 + 5x - 6 + x^2 - 6x - 27 = 2x^2 - 7x + 3 \checkmark$$

$$6x = 36$$

$$x = 6 \checkmark$$

b) $x^2 + 5x + \frac{6}{x^2+5x} = 7$ (10)

Let $k = x^2 + 5x$ ✓

Rest
 $k \neq 0$ or $x \neq -5$ ✓
 $x \neq 0$

$$k + \frac{6}{k} = 7$$

$$k^2 + 6 = 7k \checkmark$$

$$k^2 - 7k + 6 = 0$$

$$(k-6)(k-1) = 0 \checkmark$$

$$x^2 + 5x - 6 = 0 \checkmark \quad \text{or} \quad x^2 + 5x - 1 = 0 \checkmark$$

$$(x+6)(x-1) = 0$$

$$x = -6 \checkmark \quad \text{or} \quad x = 1 \checkmark$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-5 \pm \sqrt{25 + 4}}{2} \checkmark$$

$$x = \frac{-5 \pm \sqrt{29}}{2} \checkmark$$

c) $(x-2)^2(x+1) = (x+1)$

(8)

$$(x-2)^2(x+1) - (x+1) = 0$$

Let $k = x+1$ ✓

$$(k-3)^2 k - k = 0$$
 ✓

$$(k^2 - 6k + 9)k - k = 0$$

$$k(k^2 - 6k + 9 - 1) = 0$$

$$k(k^2 - 6k + 8) = 0$$
 ✓

$$k(k-2)(k-4) = 0$$
 ✓

$$k = 0 \quad \text{or} \quad k = 2 \quad \text{or} \quad k = 4$$

$$x+1 = 0 \quad \text{or} \quad x+1 = 2 \quad \text{or} \quad x+1 = 4$$

$$x = -1 \quad \text{or} \quad x = 1 \quad \text{or} \quad x = 3$$
 ✓

$$(x-1)[(x-2)^2 - 1] = 0$$
 ✓

$$(x-1)(x^2 - 4x + 4 - 1) = 0$$
 ✓

$$(x-1)(x^2 - 4x + 3) = 0$$
 ✓

$$(x-1)(x-3)(x-1) = 0$$
 ✓

$$x = 1 \quad \text{or} \quad x = 3 \quad \text{or} \quad x = 1$$
 ✓

OR

QUESTION 2 – SOLVE FOR x

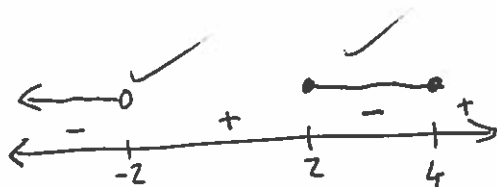
28 MARKS

NOTE: ($x \in \mathbb{R}$)

a) $\frac{x^2 - 6x + 8}{x+2} \leq 0$

(5)

$$\frac{(x-2)(x-4)}{x+2} \leq 0$$
 ✓



$$x < -2 \quad \text{or} \quad 2 \leq x \leq 4$$
 ✓

b) $\frac{x^2-20x+100}{|x+80|} \geq -50$

(3)

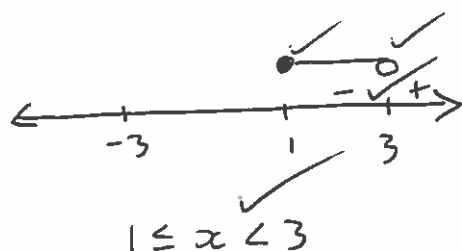
$$\frac{(x-10)^2}{|x+80|} \geq -50$$

$$x \in \mathbb{R} \checkmark \checkmark$$

c) $\frac{\sqrt{x-1}}{x^2-9} \leq 0$

(6)

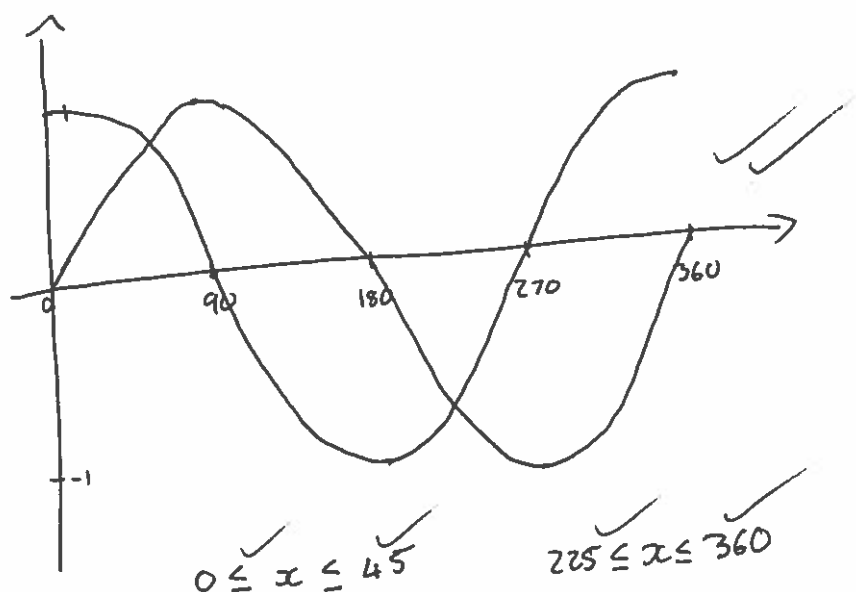
$$\frac{\sqrt{x-1}}{(x-3)(x+3)} \leq 0$$



d) $\sin x - \cos x \leq 0$ where $x \in [0^\circ; 360^\circ]$

(6)

$$\sin x \leq \cos x$$



e) Let $f(x) = \frac{\sqrt{x+8}}{|x|+\sqrt{8}}$

i) Is it possible for x to be negative? Give a reason for your answer.

(3)

Yes, ✓ counter example: $x = -1$

$$f(-1) = \frac{\sqrt{-1+8}}{|-1|+\sqrt{8}} \checkmark \checkmark$$

$$= \frac{\sqrt{7}}{1+\sqrt{8}}$$

ii) Is it possible for $f(x)$ to be negative? Give a reason for your answer.

(3)

No, ✓ $\sqrt{x+8} \geq 0$ ✓ and $|x| \geq 0$; $\sqrt{8} \geq 0$ ✓

iii) State the Domain of this function.

(2)

$$x \geq -8 \checkmark$$

QUESTION 3 – SOLVE FOR x

12 MARKS

NOTE: ($x \in \mathbb{C}$)

YOU MAY NOT USE A CALCULATOR FOR THE FOLLOWING QUESTIONS

a) $x^8 - 8x^6 = 20x^4$ (6)

$$x^8 - 8x^6 - 20x^4 = 0$$

$$x^4(x^4 - 8x^2 - 20) = 0$$

$$x^4(x^2 - 10)(x^2 + 2) = 0$$

$$x = 0 \quad \text{or} \quad x = \pm\sqrt{10} \quad \text{or} \quad x = \pm\sqrt{2}i$$

b) $x^3 - x^2 + 4x + 170 = 0$, given that $x = 3 - 5i$ is a solution (6)

$$x = 3 + 5i \quad \text{is also a solution.}$$

$$\begin{aligned} x^2 - 2ax + (a^2 + b^2) &= x^2 - 2(3)x + (3^2 + 5^2) \\ &= x^2 - 6x + 34 \end{aligned}$$

$$(x^2 - 6x + 34)(x - k) = x^3 - x^2 + 4x + 170$$

$$(x^2 - 6x + 34)(x + 5) = x^3 - x^2 + 4x + 170$$

$$x = 3 \pm 5i \quad \text{or} \quad x = -5$$

QUESTION 4 – SOLVE FOR x

NOTE: ($x \in \mathbb{C}$)

29 MARKS

a) $2|x^2 - 5x + 9| = 6$

(8)

$$|x^2 - 5x + 9| = 3 \quad \checkmark$$

$$x^2 - 5x + 9 = 3$$

or

$$x^2 - 5x + 9 = -3 \quad \checkmark$$

$$x^2 - 5x + 6 = 0$$

or

$$x^2 - 5x + 12 = 0$$

$$(x - 2)(x - 3) = 0 \quad \checkmark$$

or

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = 2 \quad \checkmark \quad \text{or} \quad x = 3 \quad \checkmark$$

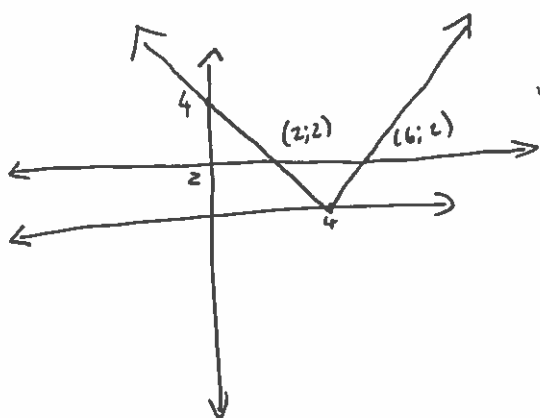
$$x = \frac{5 \pm \sqrt{25 - 48}}{2} \quad \checkmark \checkmark$$

$$x = \frac{5 \pm \sqrt{-23}}{2}$$

$$x = \frac{5 \pm \sqrt{23}i}{2} \quad \checkmark \checkmark$$

b) $|x - 4| \leq 2$

(4)



$$2 \leq x \leq 6 \quad \checkmark$$

OR

$$x - 4 \leq 2 \quad \checkmark$$

$$x \leq 6 \quad \checkmark$$

or

$$x - 4 \geq -2 \quad \checkmark$$

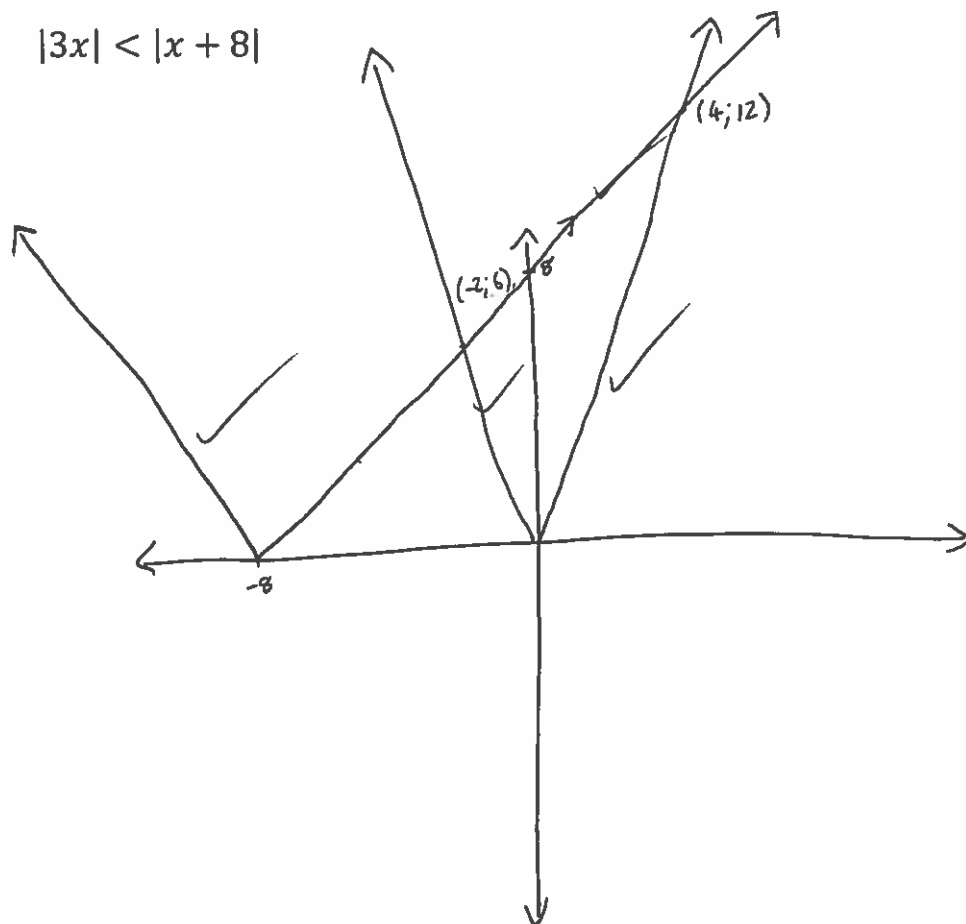
or

$$x \geq 2 \quad \checkmark$$

$$2 \leq x \leq 6 \quad \checkmark$$

c) $|3x| < |x+8|$

(8)



$$3x = x + 8$$

$$3x - x = 8$$

$$2x = 8$$

$$x = 4 \checkmark$$

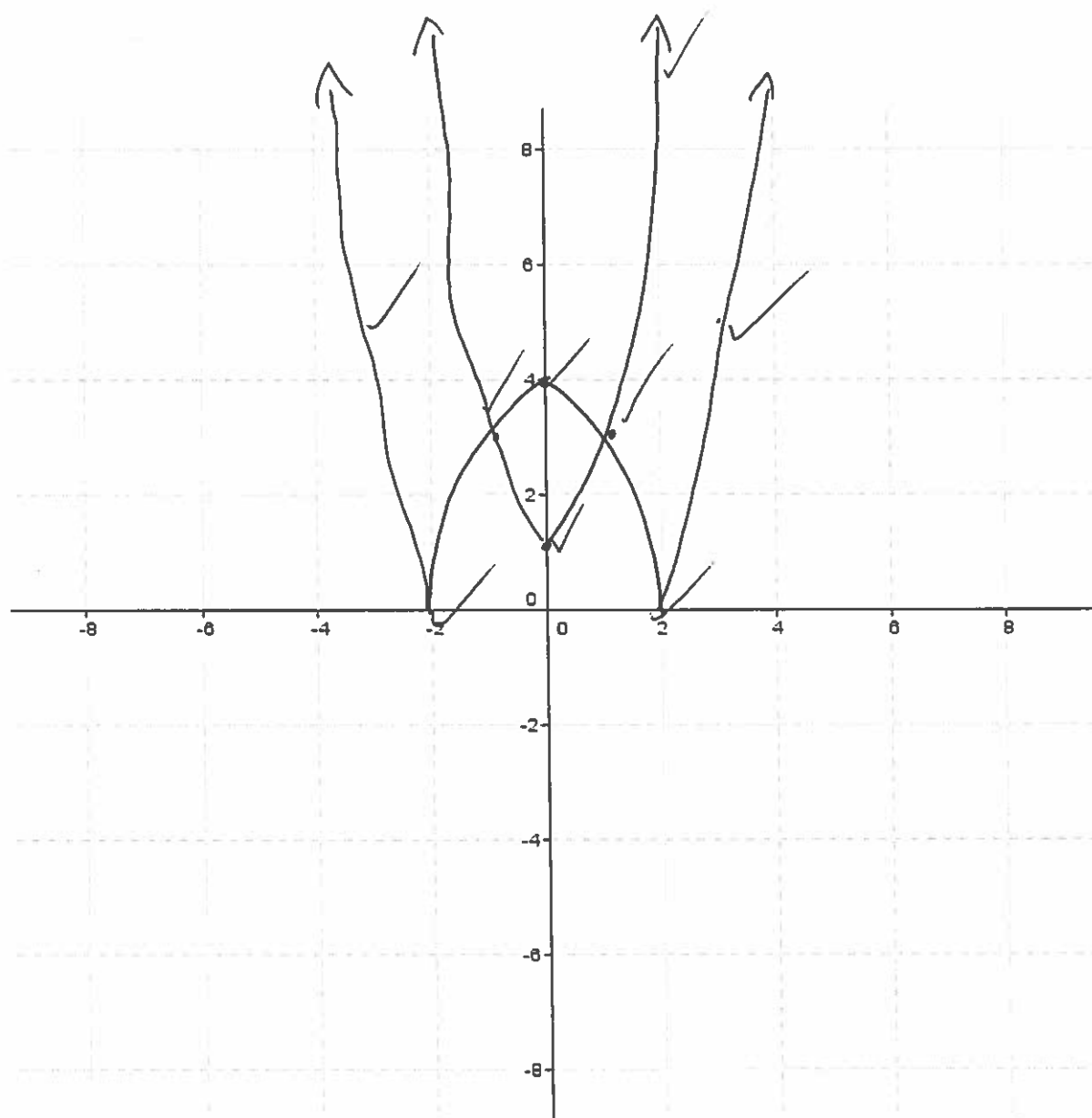
$$-3x = x + 8$$

$$-4x = 8$$

$$x = -2 \checkmark$$

$$-2 \leq x \leq 4 \checkmark \checkmark$$

- d) Sketch the graph of $f(x) = |x^2 - 4|$ and $g(x) = 3^{|x|}$ on the following set of axis showing all important points. (9)



QUESTION 5

24 MARKS

a) Simplify the following expression as far as possible:

(8)

$$\frac{\sec(270+\theta) \cdot \cos(90-\theta)}{\cos^2(180+\theta)} + \frac{\cot(270+\theta)}{\cot(\theta)}$$

$$\frac{\csc \theta \cdot \sin \theta}{\cos^2 \theta} - \frac{\tan \theta}{\cot \theta}$$

$$= \frac{1 \cdot \sin \theta}{\sin \theta \cdot \cos^2 \theta} - \frac{\sin \theta}{\cos \theta} \div \frac{\cos \theta}{\sin \theta}$$

$$= \frac{1}{\cos^2 \theta} - \frac{\sin^2 \theta}{\cos^2 \theta}$$

$$= \frac{1 - \sin^2 \theta}{\cos^2 \theta}$$

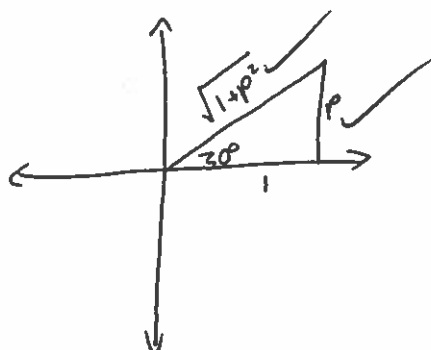
$$= \frac{\cos^2 \theta}{\cos^2 \theta}$$

$$= 1$$

b) Let $\tan 30^\circ = p$,

i) Sketch a diagram that can be used to find the following in terms of p :

(2)



ii) $\cot 30^\circ$

(2)

$$= \frac{1}{\tan 30} = \frac{1}{p}$$

ii) $\sec 150^\circ$

(2)

$$\begin{aligned} &= -\sec 30 \\ &= -\frac{\text{hyp}}{\text{adj}} \\ &= -\frac{\sqrt{1+p^2}}{1} \end{aligned}$$

ii) $\operatorname{cosec} 120^\circ$

(3)

$$\begin{aligned} &= \operatorname{cosec} 60 \\ &= \sec 30 \\ &= \sqrt{1+p^2} \end{aligned}$$

c) Prove: $\frac{\cos x}{\operatorname{cosec}^2 x - \cot^2 x - \sin x} - \tan x = \sec x$

(7)

$$\text{LHS} = \frac{\cos x}{\operatorname{cosec}^2 x - \cot^2 x - \sin x} - \tan x$$

$$= \frac{\cos x}{1 - \sin x} - \tan x$$

$$= \frac{\cos x}{1 - \sin x} - \frac{\sin x}{\cos x} \checkmark$$

$$= \frac{\cos^2 x - \sin x(1 - \sin x)}{(1 - \sin x)(\cos x)} \checkmark$$

$$= \frac{\cos^2 x - \sin x + \sin^2 x}{(1 - \sin x)(\cos x)}$$

$$= \frac{1 - \sin x}{(1 - \sin x)(\cos x)}$$

$$= \frac{1}{\cos x} \checkmark$$

$$= \sec x = \text{RHS} \checkmark$$

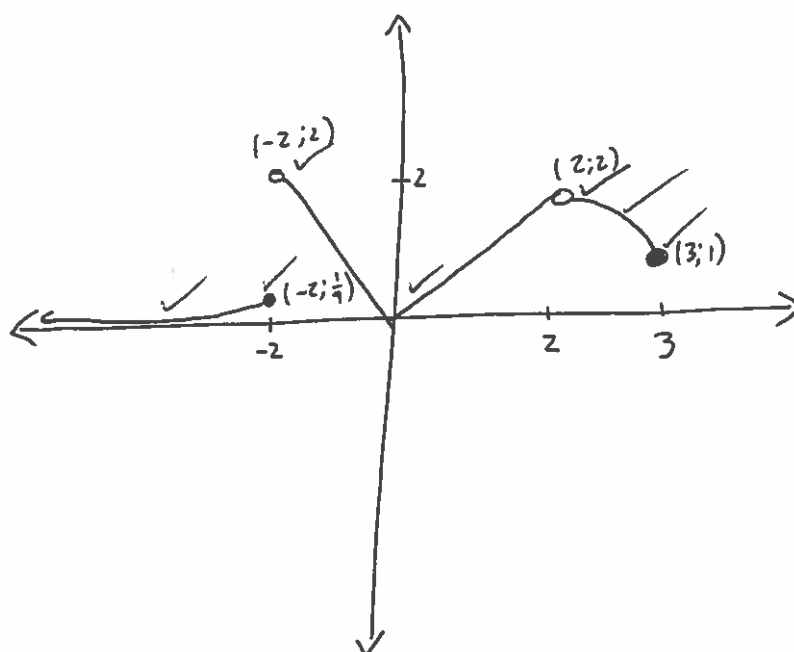
QUESTION 6

14 MARKS

Given the function: $f(x) = \begin{cases} 3^x; & x \leq -2 \\ |x|; & -2 < x < 2 \\ -x^2 + 4x - 2; & 2 < x \leq 3 \end{cases}$

a) Sketch $f(x)$

(7)



b) What is the range of this function

(2)

$$0 < y < 2$$

c) What are the zeroes of this function?

(1)

$$x=0 \checkmark$$

d) What is $f(-2)$?

(2)

$$f(-2) = 3^{-2} \checkmark$$

$$f(-2) = \frac{1}{9} \checkmark$$

e) What is $f(2)$?

(1)

Undefined $\checkmark$

f) When x gets close to 2, what does $f(x)$ get close to?

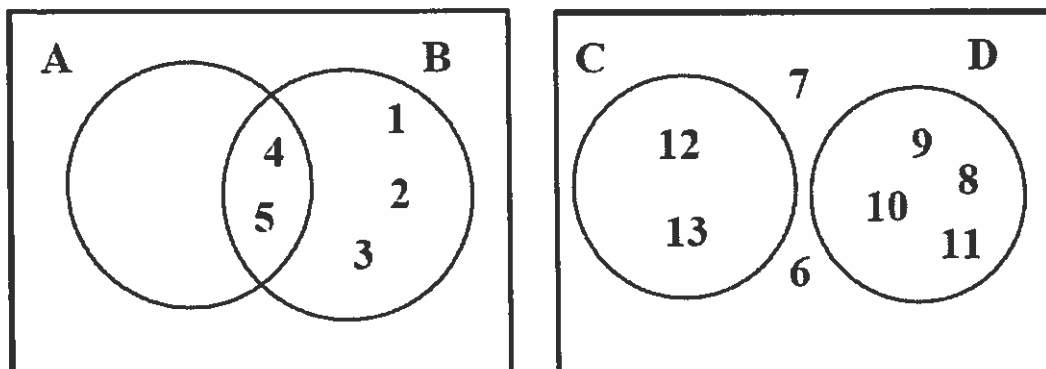
(1)

2 $\checkmark$

QUESTION 7

31 MARKS

Consider the following Venn diagrams. (Note: they are two separate diagrams and not related in any way)



a) Calculate the following:

i) $P(A) = \frac{2}{5}$ ✓ (1)

ii) the elements of set A or B (1)

$$\{1; 2; 3; 4; 5\}$$

iii) $P(C \text{ or } D)$ (1)

$$= \frac{6}{8}$$

iv) $P(B^c) = \frac{0}{5} = 0$ ✓ (1)

v) Are any of the Venn Diagrams exhaustive? Give a reason for your answer. (2)

$$P(A \cup B) = 1, \text{ Yes } A \text{ and } B$$

vi) Are any of the Venn Diagrams mutually exclusive? Give a reason for your answer. (2)

$$\text{Yes } C \text{ and } D, P(C \cap D) = 0$$

- vii) Are any of the Venn Diagrams complementary? Give a reason for your answer. (2)

No, none mutually exclusive and exhaustive

- viii) Are sets A and B independent? Give a reason for your answer. (4)

$$P(A \cap B) = \frac{2}{5} \checkmark$$

$$P(A) \times P(B) = \frac{2}{5} \times \frac{5}{5} = \frac{2}{5} \checkmark$$

$\therefore$ Independent

$$P(A \cap B) = P(A) \times P(B) \checkmark$$

- b) A Grade 11 subject survey involving Maths, AP Maths, and Science was conducted at St. Benedicts. 140 Grade 11's were surveyed and the findings were as follows:

If a learner takes AP Maths as a subject, he automatically takes Maths as a subject.

102 learners take Maths as a subject

80 Learners take Science as a subject

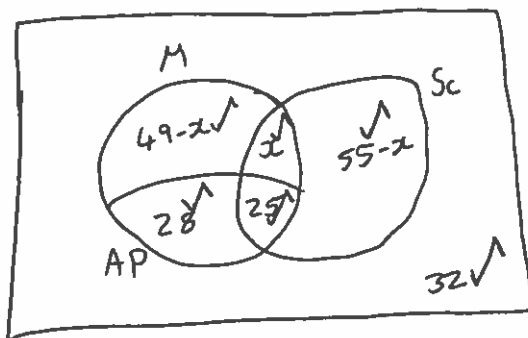
53 Learners take AP Maths as a subject

25 Learners take all three subjects.

32 Learners take none of these subjects.

Let x Learners take Maths and Science, but not AP Maths.

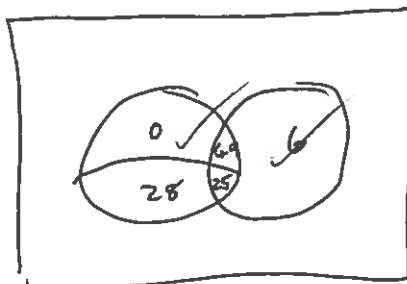
- a) Draw a Venn diagram illustrating the survey and solve for x (6)



$$49 - x + 28 + 25 + x + 55 - x + 32 = 140$$

$$-x = -49$$

$$x = 49 \checkmark$$



b) Calculate the probability that a learner at random:

i) Takes Maths and Science as a subject.

(2)

$$= \frac{74}{140}$$

ii) Takes Maths, but not Science as a subject.

(2)

$$= \frac{28}{140}$$

iii) Takes AP Maths or Science as a subject.

(3)

$$= \frac{108}{140}$$

iv) Takes AP maths given that the learner takes Science as a subject.

(4)

$$P(AP|Sc) = \frac{P(AP \cap Sc)}{P(Sc)} = \frac{25}{80}$$

QUESTION 8

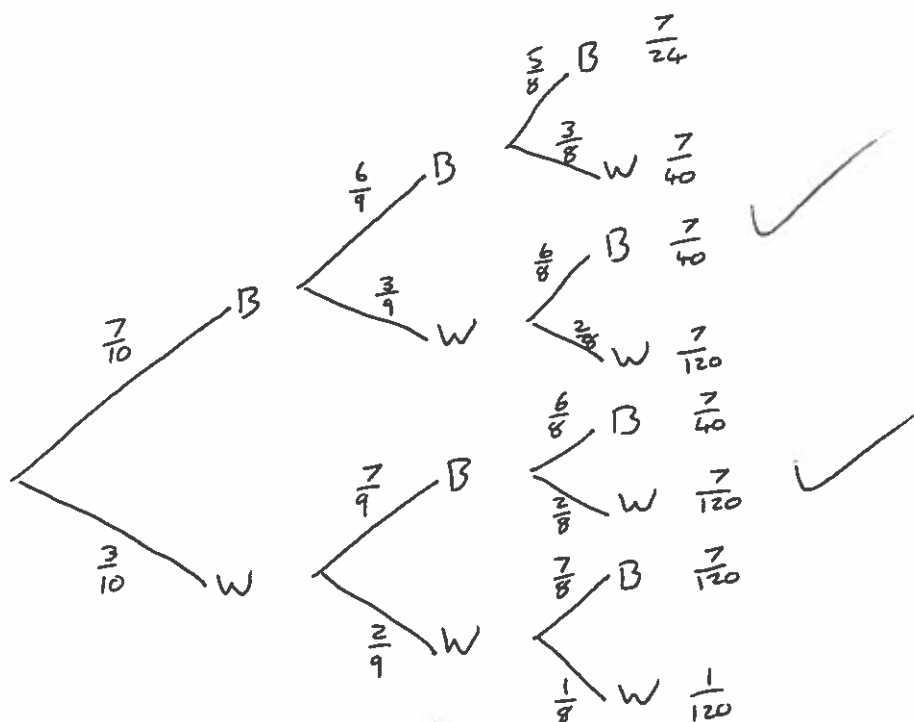
20 MARKS

- a) A jar contains 10 marbles, 7 black and 3 white. Three marbles are drawn **without** replacement.

What is the probability that:

- i) Two marbles are white?

(4)



$$P(2B) = 3 \left(\frac{7}{120} \right) = \frac{7}{40}$$

- ii) All the marbles are white?

(2)

$$P(3W) = \frac{3}{10} \times \frac{2}{9} \times \frac{1}{8} = \frac{1}{120}$$

- iii) A white marble will be drawn, given that two black marbles have already been drawn? (2)

$$P(W|2B) = \frac{3}{8}$$

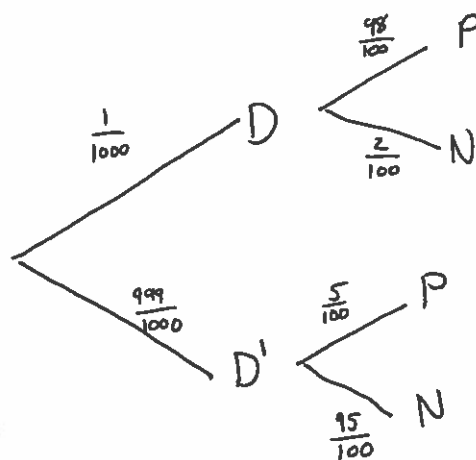
- iv) At least one black marble is drawn? (2)

$$P(\text{at least 1 B}) = 1 - P(3W) = \frac{119}{120}$$

- v) At least one black marble is drawn, given that a black marble has already been drawn? (4)

$$\left(\frac{6}{9} \times \frac{5}{8}\right) + \left(\frac{6}{9} \times \frac{3}{8}\right) + \left(\frac{3}{9} \times \frac{6}{8}\right) = \frac{11}{12}$$

- b) We know that the prevalence of a particular disease is $1/1000$ in the general population. A test for this disease has a false positive rate of 5%, that is, 5% of the time the test will incorrectly indicate that a person has the disease when in fact they do not. We are also told that 98% of the people that have the disease will in fact test positive (that they have the disease). Assuming that you know nothing about particular individuals or their symptoms, what is the probability that a person that has tested positive does in fact have the disease? (Give the probability as a fraction). (6)



$$\frac{\frac{1}{1000} \times \frac{98}{100}}{\left(\frac{1}{1000} \times \frac{98}{100}\right) + \left(\frac{999}{1000} \times \frac{5}{100}\right)} = \frac{98}{5093}$$

QUESTION 9

16 MARKS

a) Consider the word REDEEMED, where the same letter may not be used more than once.

- i) How many eight-letter word arrangements can be made if the repeated letters are treated as identical letters? (3)

$$\frac{8!}{4! \times 2!} = 840 \text{ ways}$$

- ii) What is the probability that the word starts and ends with the same letter made if the repeated letters are treated as identical letters? (6)

$$\frac{\frac{6!}{2! \times 2!} + \frac{6!}{4!}}{840} = \frac{180 + 30}{840} = \frac{1}{4}$$

b) Five boys and six girls are to be seated randomly in a row. What is the probability that:

i) Two particular girls sit next to each other

(3)

$$\frac{10! \times 2!}{11!} = \frac{2}{11}$$

ii) The boys and girls alternate?

(4)

$$6! \times 5! = 86400 \text{ ways}$$

BONUS QUESTION

The digits 0, 1, 2, 3, 4, 5, 6 and 7 are used to make a 4 digit code (digits may be repeated). What is the probability that the code will be less than or equal to 3000.

$$\underline{3} \ \underline{8} \ \underline{8} \ \underline{8} + 1 = 1537$$

$$8^4 = 4096$$

$$P(\leq 3000) = \frac{1537}{4096} \checkmark \checkmark$$